

AER1217 Lab 3: Georeferencing Using UAV Payload Data

Group 2:

Zouhair Adam Hamaimou (Student #1004891986)

Mark Zhuang (Student #1006963644)

Yi Lo (Student #1009764550)

University of Toronto Institute for Aerospace Studies

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1 Introduction

In this lab, we used the Parrot AR.Drone 2.0's downward-facing camera to detect and georeference circular targets within the Vicon tracking volume. Our objectives were to apply the pinhole camera model, use color-based segmentation to detect targets, transform 2D detections into 3D Vicon coordinates, and refine positions using clustering and outlier rejection.

2 Algorithm and Discussion

2.1 Camera Model and Pose Composition

To map image detections to 3D space, we employed the **pinhole camera model** and pose composition.

2.1.1 Lens Distortion Removal

The image undergoes distortion correction using OpenCV's `undistort` function with known radial and tangential distortion coefficients:

$$\mathbf{d} = [0.191887, -0.563680, -0.003676, -0.002037, 0.0]$$

2.1.2 Intrinsic and Extrinsic Transformations

Each pixel coordinate $\mathbf{u} = (u, v)$ is mapped to a normalized camera-frame ray:

$$\mathbf{r}_c = \mathbf{K}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 698.86 & 0.0 & 306.91 \\ 0.0 & 699.13 & 150.34 \\ 0.0 & 0.0 & 1.0 \end{bmatrix}$$

The camera ray is transformed to the inertial frame via:

$$\mathbf{T}_{IC} = \mathbf{T}_{IB} \mathbf{T}_{CB}^{-1} = \begin{bmatrix} \mathbf{R}_{IC} & \mathbf{t}_{IC} \\ \mathbf{0} & 1 \end{bmatrix}$$

where $\mathbf{T}_{IB}$ represents the drone's position (p_x, p_y, p_z) and orientation (q_w, q_x, q_y, q_z) , converted to a rotation matrix.

The body-to-camera transformation is given by:

$$\mathbf{T}_{CB} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Since $\mathbf{T}_{CB}$ is a pure rotation and symmetric, its inverse simplifies to:

$$\mathbf{T}_{CB}^{-1} = \mathbf{T}_{CB}^T = \mathbf{T}_{CB}$$

allowing us to compute:

$$\mathbf{T}_{IC} = \mathbf{T}_{IB}\mathbf{T}_{CB}$$

The full homogeneous transformation from the inertial to the camera frame is:

$$\mathbf{T}_{IB} = \begin{bmatrix} \mathbf{R}_{IB} & \mathbf{t}_{IB} \\ \mathbf{0} & 1 \end{bmatrix}$$

where $\mathbf{t}_{IB} = (p_x, p_y, p_z)^T$ is the translation vector.

2.2 Ray-Ground Intersection

The transformed camera ray is intersected with the ground plane ($z = 0$) to compute 3D world coordinates. The ray equation in the inertial frame is:

$$\mathbf{p}(t) = \mathbf{c}_I + t\mathbf{d}_I$$

where $\mathbf{c}_I = \mathbf{t}_{IC} = (c_x, c_y, c_z)$ is the camera center, and $\mathbf{d}_I = \mathbf{R}_{IC}\mathbf{r}_c = (d_x, d_y, d_z)$ is the ray direction.

Solving for t at $z = 0$:

$$t = -\frac{c_z}{d_z}, \quad X = c_x + td_x, \quad Y = c_y + td_y$$

These coordinates represent the projected 2D ground points.

2.3 Target Detection and Clustering

Green targets are detected using HSV-based segmentation with thresholds:

$$\text{Lower: } (30, 60, 20), \quad \text{Upper: } (80, 255, 95)$$

Morphological filtering (kernel size = 5) is applied to remove noise. Detections are validated based on:

- **Minimum area:** $A_{\min} = 200$ pixels.
- **Circularity:** $A_{\text{contour}}/A_{\text{circle}} \geq 0.5$, where:
 - A_{contour} is the detected contour area.
 - $A_{\text{circle}} = \pi r^2$ is the enclosing circle area.

The detected centers are transformed to inertial coordinates using Section 2.1.2 and projected onto the ground via Section 2.2.

2.4 Robust Position Estimation

2.4.1 K-Means Clustering

Since there are six ground targets, detections are grouped using K-Means clustering with $k = 6$. Given N observations $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$, each point is assigned to a cluster center μ_j such that:

$$\arg \min_{\mu_j} \sum_{i=1}^N \|\mathbf{X}_i - \mu_j\|^2$$

where $\|\cdot\|$ denotes Euclidean distance.

2.4.2 RANSAC-Based Outlier Rejection

Within each cluster, the target position is refined using RANSAC, which iteratively:

1. Selects a random observation as the seed.
2. Identifies inliers within a threshold distance ϵ .
3. Updates the estimated position using the inlier mean.

Formally, for a cluster with points $\mathbf{X}$, a point $\mathbf{x}_i$ is considered an inlier if:

$$\|\mathbf{x}_i - \mathbf{x}_{\text{seed}}\| < \epsilon$$

The final estimated target position is obtained by averaging the inliers:

$$\hat{\mathbf{x}} = \frac{1}{|\mathcal{I}|} \sum_{\mathbf{x}_i \in \mathcal{I}} \mathbf{x}_i$$

where $\mathcal{I}$ is the set of inliers.

2.5 Final Target Estimation

After processing multiple frames, the refined cluster centers represent the final estimated positions of the six targets. These positions are stored for further analysis.

3 Results and Discussion

The final (X, Y) coordinates of the six detected targets were computed using the complete processing pipeline, incorporating clustering and RANSAC for robust estimation. The refined target positions are presented in Table 1.

3.1 Multi-Frame Aggregation for Improved Accuracy

To enhance localization robustness, target detection was performed across multiple frames, incorporating observations from different viewpoints. This multi-frame approach mitigates errors due to occlusions, perspective distortions, and sensor noise. The final reported positions represent the centroid of each detected circle projected onto the floor plane, expressed in Vicon coordinates.

Table 1: Final Target Localizations (RANSAC + Clustering)

Target	X Coordinate	Y Coordinate
1	0.330344088	-0.267532431
2	-1.023600271	0.969885010
3	-1.027216498	-0.706050464
4	0.051080989	-1.926232190
5	1.432164658	-1.470353848
6	0.631355243	0.855592423

3.2 Visualization of Detected and Final Positions

The scatter plot in Figure 1 illustrates the detected positions (blue) and the final estimated target locations (red crosses) after applying clustering and RANSAC.

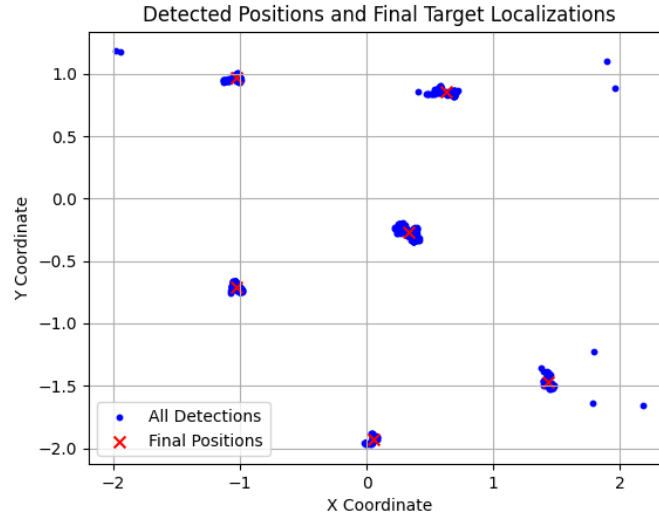


Figure 1: Detected Positions and Final Target Localizations using Clustering and RANSAC. The blue points represent all detected positions across multiple frames, while the red crosses indicate the final estimated positions of the six targets.

4 Conclusion

We detected six green circular targets, transformed them into inertial coordinates, and refined their locations using k -means clustering and 2D RANSAC. Our approach followed standard computer vision and rigid-body transformations, including distortion removal, pixel-to-ray conversion, pose composition, and ray-plane intersection. The results demonstrate that multiple observations and robust estimation improve target localization.